

Classical Mechanics

Lecture 1: States, Reversibility, and the Order of Motion

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Chapter 1

States, Reversibility, and the Order of Motion

Classical mechanics is the foundation on which the rest of physics is built. This is not merely because it describes particles, planets, and mechanical devices. Its deeper contribution is a framework for asking what a physical state is, how a state changes, what information a law of motion must preserve, and why quantities such as energy and momentum can remain constant. The same questions survive in more abstract form throughout modern physics.

We shall not begin with a falling body or with Newton's equation. Instead, we shall strip dynamics down to its logical skeleton. Once that skeleton is visible in a world with only a few possible states, the passage to continuous motion will be much easier to understand.

1.1 A Stroboscopic World

Ordinary time appears continuous: the second hand of a watch sweeps through every intermediate position. For the moment, imagine instead that the world is illuminated by a strobe. We inspect it only at equally spaced beats,

$$t_n = n\Delta t, \quad n \in \mathbb{Z}, \quad (1.1)$$

where Δt may be a second, a millisecond, or any other fixed interval. A law of motion is then an update rule

$$q_n \mapsto q_{n+1}. \quad (1.2)$$

The question “What happens next?” has become literal: given the state at one flash, what will be seen at the following flash?

We also choose systems much simpler than anything in nature. Their purpose is not realism but clarity. The simplest nontrivial example is a coin laid on a table. It has two possible configurations, heads and tails, which we denote by H and T . Its space of possibilities is

$$\Gamma = \{H, T\}. \quad (1.3)$$

Definition 1.1. The *phase space* is the set of all possible states of a system. A *state* is a complete specification of the information needed to determine what happens next.

This definition is operational. A description is not a state merely because we call it one. If it does not contain enough information to determine the next step, it is incomplete and the phase space must be enlarged.

A world with only one state would be still simpler, but nothing could happen there. Its only possible law would send that state back to itself forever. Two states provide the first opportunity for a genuine alternative.

1.2 Two Elementary Laws

One possible law leaves the coin unchanged:

$$H \mapsto H, \quad T \mapsto T. \quad (1.4)$$

We represent the rule by drawing a loop from each state back to itself. Starting at the tail of an arrow and following it to its head tells us the next state.

A second law flips the state at every beat:

$$H \mapsto T, \quad T \mapsto H. \quad (1.5)$$

Starting from heads, the complete history is

$$H, T, H, T, H, T, \dots \quad (1.6)$$

Both laws are deterministic. Once the state at one instant is known, the state at every later instant is fixed.

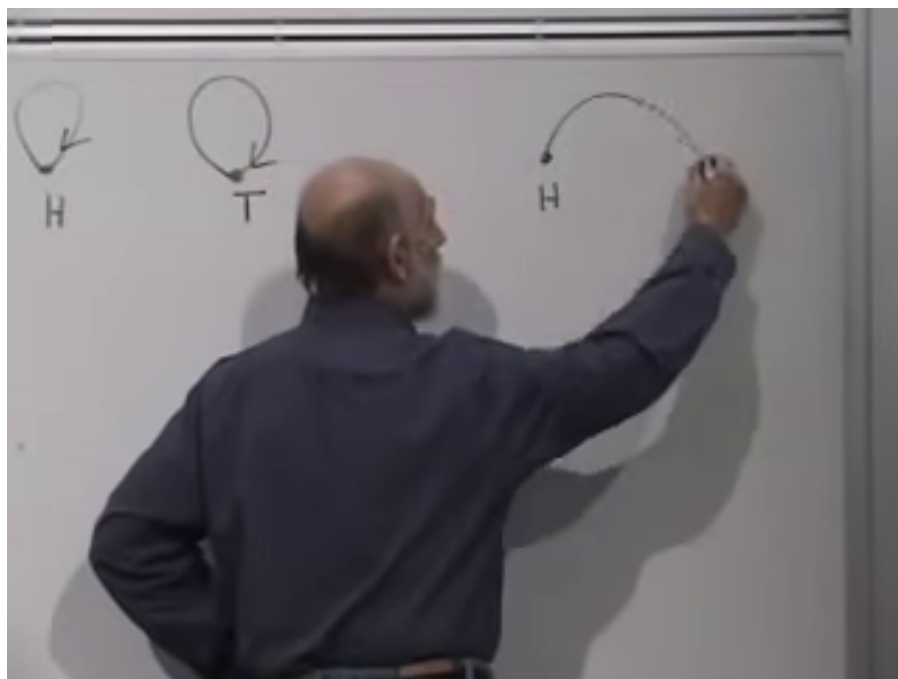


Figure 1.1: The board at 00:06:58. The two self-loops are complete, and the first cross-arrow of the alternating law is being drawn. The reverse arrow is supplied by the spoken rule, not inferred from the unfinished chalk mark.

Lecture frame: 00:06:58.



Figure 1.2: The two reversible laws on a two-state phase space.

Question from the audience. Does determinism already imply that the law is reversible?^a

Response. Not by itself. Determinism says that the present has a unique future. Reversibility requires the converse as well: the present must have a unique past. The two laws above happen to satisfy both conditions, but a larger state space lets us separate them.

^aLecture timestamp: 00:06:43–00:06:50.

1.3 More States and More Orbits

Replace the coin by a die. A die has six states, labelled $1, 2, \dots, 6$. One possible law is the six-cycle

$$1 \mapsto 2 \mapsto 3 \mapsto 4 \mapsto 5 \mapsto 6 \mapsto 1. \quad (1.7)$$

The physical tumbling needed to display these faces might look complicated, but the abstract law is simple: every state has a prescribed successor.

The numerical order is inessential. We could relabel the same cycle, or divide the states into several disjoint cycles. For example,

$$1 \mapsto 1, \quad 2 \mapsto 3 \mapsto 4 \mapsto 2, \quad 5 \mapsto 6 \mapsto 5. \quad (1.8)$$

This law contains a fixed point, a three-cycle, and a two-cycle. Wherever the system starts, the arrow diagram determines its future indefinitely. It also determines its past: every state has exactly one predecessor.

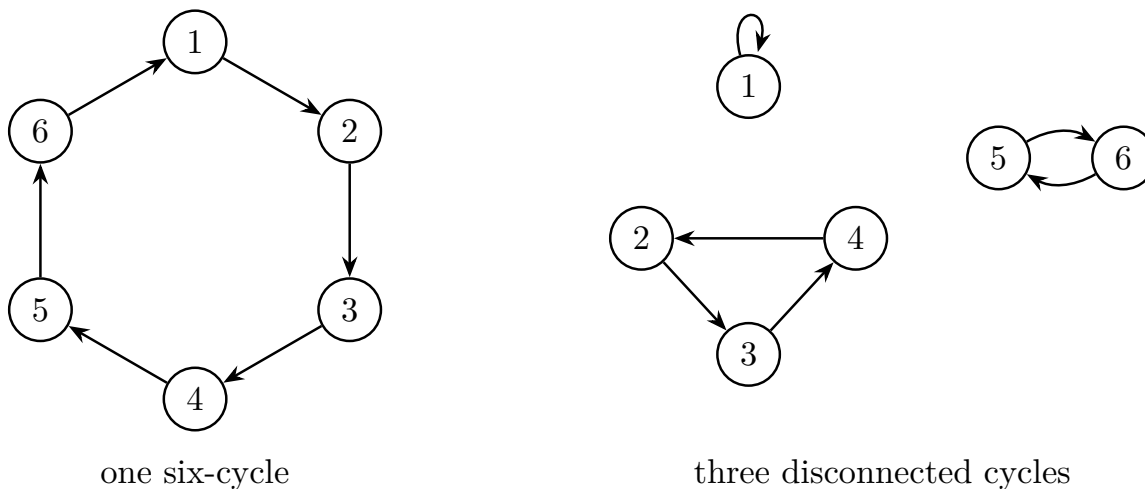


Figure 1.3: A finite reversible law is built from disjoint cycles, including cycles of length one.

In practice, a complicated system may be so sensitive to small errors that reconstructing its history is hopeless. That practical difficulty does not alter the ideal law. If the state were known exactly, its trajectory could be followed both forward and backward.

1.4 Forbidden Maps

Consider three states with the rule

$$1 \mapsto 2, \quad 2 \mapsto 3, \quad 3 \mapsto 2. \quad (1.9)$$

The future is unique. From any state there is one arrow to follow. The past is not unique, however. If the system is now at 2, it may have come from 1 or from 3. Two distinct histories have merged.

Reversing every arrow exchanges the defect. The past then becomes unique, but state 2 has two possible futures. Nothing in the state tells us whether to choose 1 or 3.

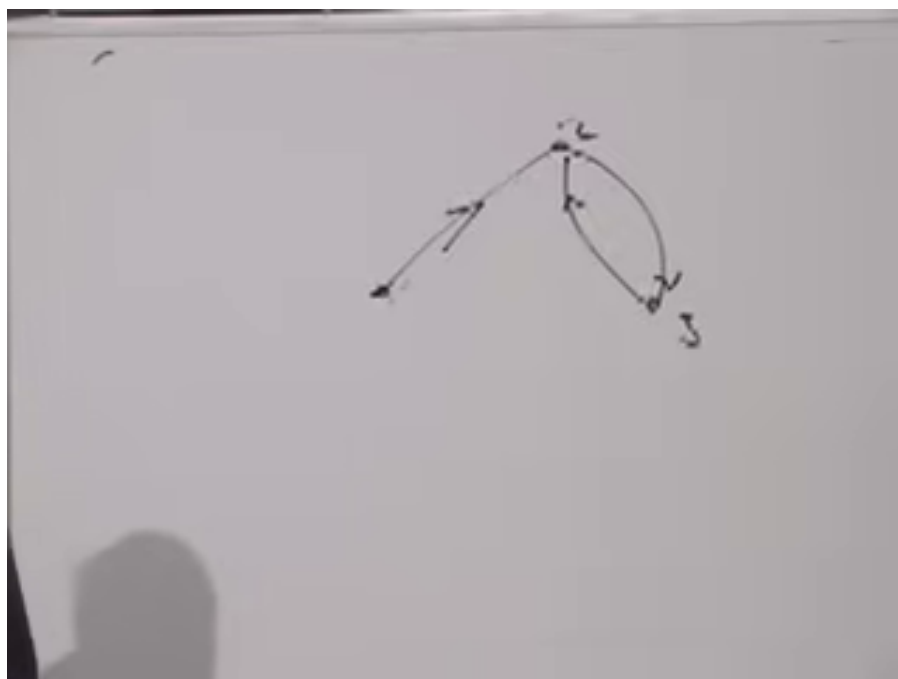


Figure 1.4: The board at 00:12:46, showing the merging three-state map. The two arrows entering state 2 make its past ambiguous.

Lecture frame: 00:12:46.

Question from the audience. Could we repair one of these diagrams by erasing an arrow, or by letting a chain terminate at a state?^a

Response. Erasing an outgoing arrow leaves a state with no prescribed future. Turning the endpoint into a fixed point repairs the future but generally spoils the past: the fixed point then has both its self-loop and an incoming arrow from the chain. One way or another, a missing or dangling arrow violates uniqueness.

^aLecture timestamp: 00:14:46–00:15:34.

The criterion can now be stated exactly.

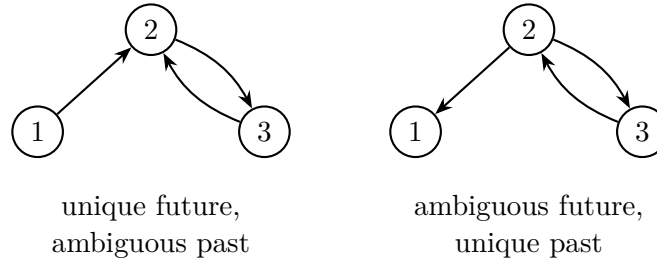


Figure 1.5: The two ways a state map can fail the classical one-in, one-out condition.

Proposition 1.2. *For a finite phase space Γ , a reversible deterministic update is a bijection*

$$\sigma : \Gamma \longrightarrow \Gamma. \quad (1.10)$$

Equivalently, every state has exactly one outgoing arrow and exactly one incoming arrow.

Proof. One outgoing arrow gives a unique future, and one incoming arrow gives a unique past. If either arrow is missing, evolution stops in one time direction. If either is duplicated, evolution is ambiguous in that direction. On a finite set, a map with one incoming and one outgoing arrow at every point is a permutation, hence a disjoint union of cycles. $\square$

The inevitable cycling of a finite system may seem dull, but it is a consequence of finiteness, not of reversibility. An infinite phase space need not recur. If states are labelled by every integer and

$$n \mapsto n + 1, \quad n \in \mathbb{Z}, \quad (1.11)$$

then each state still has one predecessor and one successor, while the trajectory marches along the line forever.

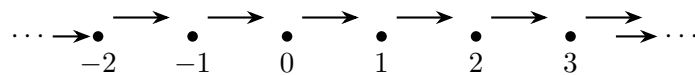


Figure 1.6: A reversible law on an infinite phase space need not return to its starting point.

1.5 Conservation as Memory

Phase space may split into disconnected components. Some initial states may lie on an infinite chain, while others lie on closed cycles. Once the system starts in one component, it can never reach another. That persistent distinction is the simplest model of a conservation law.

Assign a label C to each component, for example

$$C(q) = \begin{cases} +1, & q \in \Gamma_+, \\ -1, & q \in \Gamma_-. \end{cases} \quad (1.12)$$

Since the dynamics never crosses between Γ_+ and Γ_- ,

$$C(q_{n+1}) = C(q_n). \quad (1.13)$$

The conserved quantity is a memory of where the motion began. By contrast, if the labels $+1$ and -1 are assigned to the alternating states H and T , the label changes on every beat and is not conserved.

Energy and momentum are much richer quantities, but the logical form is already present: a conserved function is constant along each allowed trajectory.

The same idea may be called *information conservation*. In the merging map of Eq. (1.9), arriving at state 2 erases the distinction between two possible pasts. In a reversible map, no such distinction disappears. Exact knowledge of the present determines both where the system came from and where it is going.

Why should fundamental laws have this form? At this level there is no deeper derivation. It is an empirical fact about the known fundamental dynamics. Perhaps a future “law of laws” would explain why information is conserved. Quantum mechanics changes the mathematical language, but its closed-system evolution is also information preserving. Our concern here is the classical version.

The qualification *closed system* matters. Friction, diffusion, and thermodynamic irreversibility often look many-to-one because unobserved environmental details have been discarded. Such effective descriptions do not overturn the reversible microscopic framework being isolated here.

1.6 When the Visible Symbol Is Not the State

An audience question exposes the most important subtlety in the definition of state.

Question from the audience. Suppose the next coin value depends not only on the present face but on the two most recent faces. Seeing H now would not determine what comes next. Has determinism been lost?^a

Response. Not necessarily. The correct conclusion is that H or T alone was never a complete state. If two entries are needed to apply the law, the state must be the ordered pair of entries. What we call phase space must contain all the memory on which the update depends.

^aLecture timestamp: 00:22:09–00:23:18.

This observation leads directly to continuous mechanics. For a particle moving on a line, is its present position a complete state?

1.7 A Particle on a Line

Let x denote the particle’s position. Knowing x alone is plainly insufficient. Two particles can occupy the same point while moving in opposite directions, or while moving at different speeds. To predict the next position we must also know the velocity

$$v = \dot{x} = \frac{dx}{dt}. \quad (1.14)$$

Equivalently, over a short discrete interval we need both the present position and the preceding one, since their difference determines the velocity. The continuous state is therefore the pair

$$(x, v). \quad (1.15)$$

The phase space of a particle on a line is two-dimensional, even though the particle moves in one-dimensional physical space. In canonical mechanics one often uses (x, p) with $p = mv$; for fixed mass this carries the same information as (x, v) .

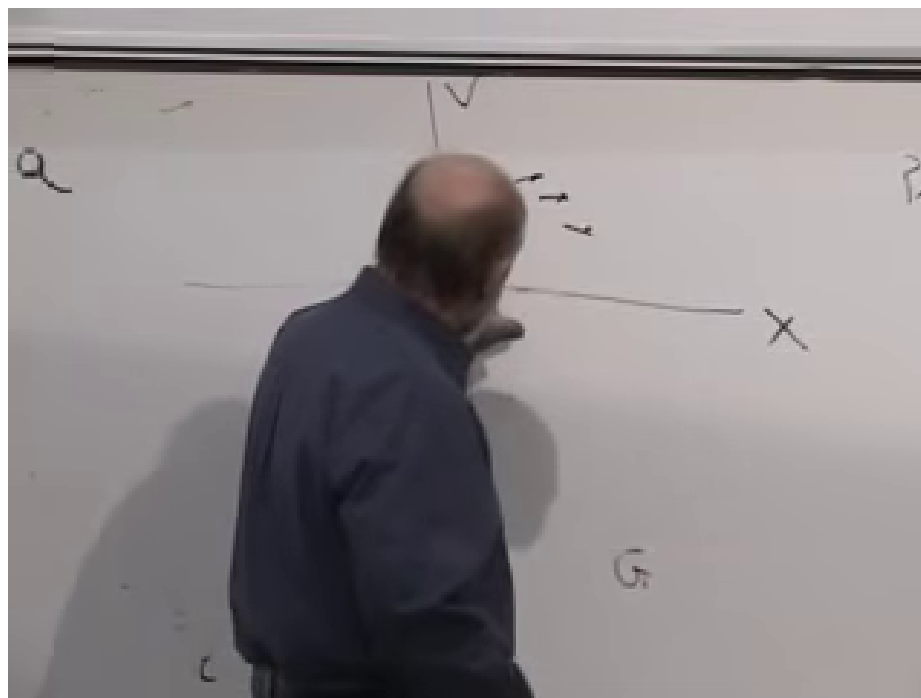


Figure 1.7: The board at 00:38:10 revisits the (x, v) phase-space plane. The horizontal x -axis and vertical v -axis are visible; the small arrows are qualitative, not a complete measured vector field.

Lecture frame: 00:38:10.

Imagine filling the (x, v) plane with short arrows. At a point with $v = 0$, the immediate horizontal displacement vanishes. At positive v , the particle moves to the right; a larger positive velocity produces a larger displacement during the same time interval. At negative v , it moves to the left. The arrow must also contain any change in velocity produced by the force, but even this elementary horizontal component shows why both coordinates are needed.

Question from the audience. If continuous motion is approached by making the time steps arbitrarily small, why did velocity appear as an additional coordinate?^a

Response. Because a rule may depend on more than one adjacent sample. In discrete language, the difference between the last two positions determines a velocity. We can either say that the rule remembers two positions or enlarge the state to include position and velocity. These are two descriptions of the same information. The phase space has one axis for the present position and another for the information carried by the previous step.

^aLecture timestamp: 00:27:09–00:28:18.

For a single particle in three-dimensional space, there are three position coordinates and three velocity coordinates, so phase space has six coordinates.

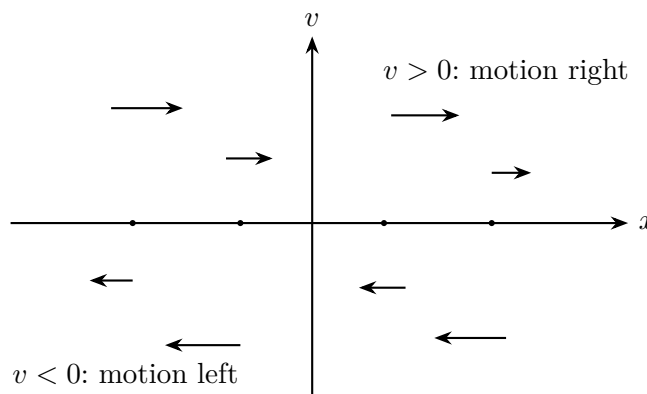


Figure 1.8: The kinematic part of the phase-space flow. The horizontal component of the arrow is set by $\dot{x} = v$.

1.8 Determinism and Practical Prediction

Classical positions and velocities are real numbers, and no experiment specifies a real number with infinite precision. Therefore exact determinism of the equations does not imply unlimited practical prediction.

Imagine demanding the position of every molecule in the room thirty seconds from now. Some finite precision in the present data would, in principle, suffice for that finite task. The same data may fail at forty seconds because its errors have had longer to grow. To extend the forecast, we must improve the initial data. Classical mechanics promises that a sufficient precision exists for each fixed interval; it does not promise that any attainable measurement remains useful forever.

Suppose an initial uncertainty $\delta z(0)$ in phase space grows approximately as

$$\delta z(t) \sim \delta z(0)e^{\lambda t}, \quad \lambda > 0. \quad (1.16)$$

Then predicting for a longer time requires exponentially more precise initial data. This is the characteristic difficulty of chaotic systems. The law still assigns one trajectory to each exact initial state, but nearby initial states separate rapidly.

Question from the audience. Can we call such a system deterministic even when it is not practically predictable?^a

Response. The distinction is useful provided it is stated carefully. The equations are deterministic: infinitely precise initial data select a unique trajectory. Actual predictions are limited because initial data are never infinitely precise. For any fixed prediction interval, classical theory says that sufficiently accurate initial data exist; extending the interval generally demands still greater accuracy. Some speakers use “deterministic” and “predictable” interchangeably, but the equations-versus-measurement distinction removes the apparent contradiction.

^aLecture timestamp: 00:28:20–00:31:55.

This is not a quantum uncertainty argument. Quantum mechanics has been deliberately set aside here. The limitation comes from finite classical measurement precision and its possible amplification by the dynamics.

1.9 The Order of the Equation

The need for both position and velocity is encoded in the order of the equations of motion. A first-order differential equation contains a first time derivative. A second-order equation contains a second derivative,

$$\dot{x} = \frac{dx}{dt}, \quad \ddot{x} = \frac{d^2x}{dt^2} = a. \quad (1.17)$$

The dot notation will be used throughout: one dot means one time derivative, two dots mean two.

To isolate the logic, consider the intentionally non-Newtonian first-order law used on the board,

$$F(x) = m\dot{x}. \quad (1.18)$$

It is a schematic law rather than a physically dimensioned Newtonian equation; a conventional first-order drag model would replace m by a coefficient with the proper units. Its pedagogical virtue is that position alone determines velocity:

$$\dot{x} = \frac{F(x)}{m}. \quad (1.19)$$

Once x is known, $F(x)$ is known, hence $\dot{x}$ is known.

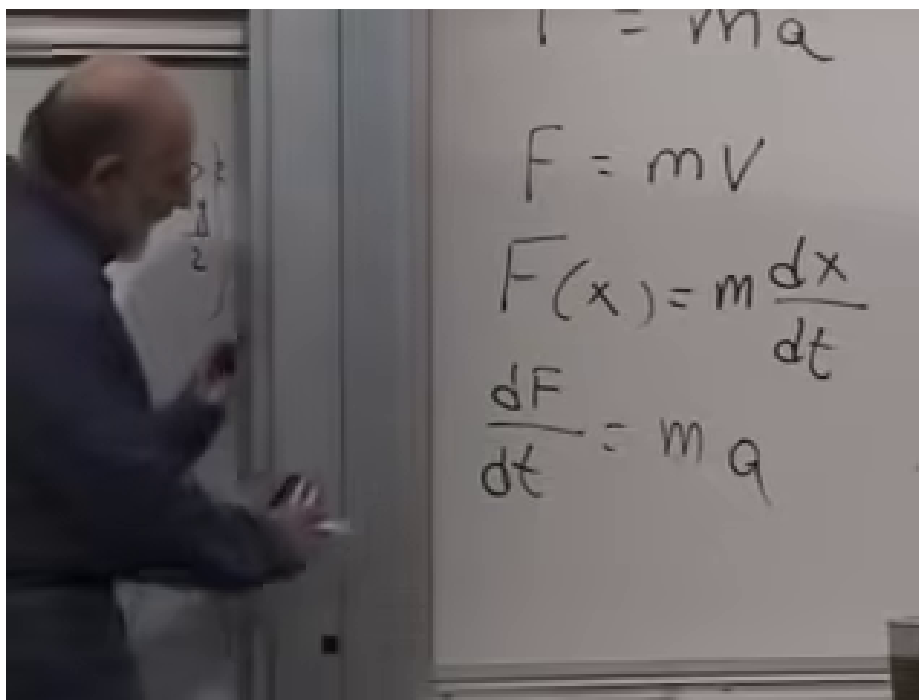


Figure 1.9: The board at 00:35:25. The deliberately phony law $F = mV$, its position-dependent form $F(x) = m dx/dt$, and the first differentiated relation are all legible.

Lecture frame: 00:35:25.

Differentiating Eq. (1.18) gives

$$\frac{dF}{dt} = m\ddot{x}, \quad (1.20)$$

$$\frac{dF}{dt} = \frac{dF}{dx} \frac{dx}{dt} = F'(x)\dot{x}. \quad (1.21)$$

Therefore

$$\ddot{x} = \frac{F'(x)}{m} \dot{x} = \frac{F'(x)F(x)}{m^2}. \quad (1.22)$$

Higher derivatives follow by differentiating again. Subject to the usual smoothness assumptions, specifying x at one time fixes the entire local Taylor expansion and hence the motion allowed by this first-order law.

Newton's actual equation is second order:

$$F(x) = m\ddot{x}. \quad (1.23)$$

Knowing x determines the force and therefore the acceleration, but it does not determine the velocity. The velocity must be supplied independently as part of the initial state. Once x and v are known, Newton's equation can be rewritten as the first-order system on phase space

$$\dot{x} = v, \quad \dot{v} = \frac{F(x)}{m}. \quad (1.24)$$

These two equations are the continuous version of the arrow leaving each phase-space point. They determine acceleration and, by further differentiation, all higher derivatives of the trajectory.

1.10 The Two-Step Coin Revisited

The discussion now returns to the coin because the discrete example makes the same point without calculus.

Question from the audience. Does the “bifurcated” heads-and-tails diagram require two pieces of information rather than one?^a

Response. No. The diagram still describes one system. The number of arrows or clauses in its law is not the number of independent pieces of state information. In Eqs. (1.4) and (1.5), one symbol— H or T —is enough to determine the next symbol. A state may have many possible values without requiring several successive values as initial data.

^aLecture timestamp: 00:38:18–00:40:41.

If the rule depends on the previous two symbols, the four possible states are

$$\Gamma_2 = \{HH, HT, TH, TT\}. \quad (1.25)$$

The state update has the shift-register form

$$(a_n, a_{n+1}) \mapsto (a_{n+1}, f(a_n, a_{n+1})). \quad (1.26)$$

The classroom builds a truth table in real time. Several first attempts fail: either the proposed output secretly depends on only one of the two entries, or two pair states flow into the same successor and information is lost. Those corrections are not incidental. They demonstrate how the one-in, one-out test detects an incomplete or irreversible rule.

One workable table is

previous pair	next symbol
HH	H
HT	T
TH	T
TT	H

(1.27)

and it induces

$$\begin{aligned} HH &\mapsto HH, & HT &\mapsto TT, \\ TT &\mapsto TH, & TH &\mapsto HT. \end{aligned} \quad (1.28)$$

Thus HH is a fixed point and the other three pair states form a cycle.

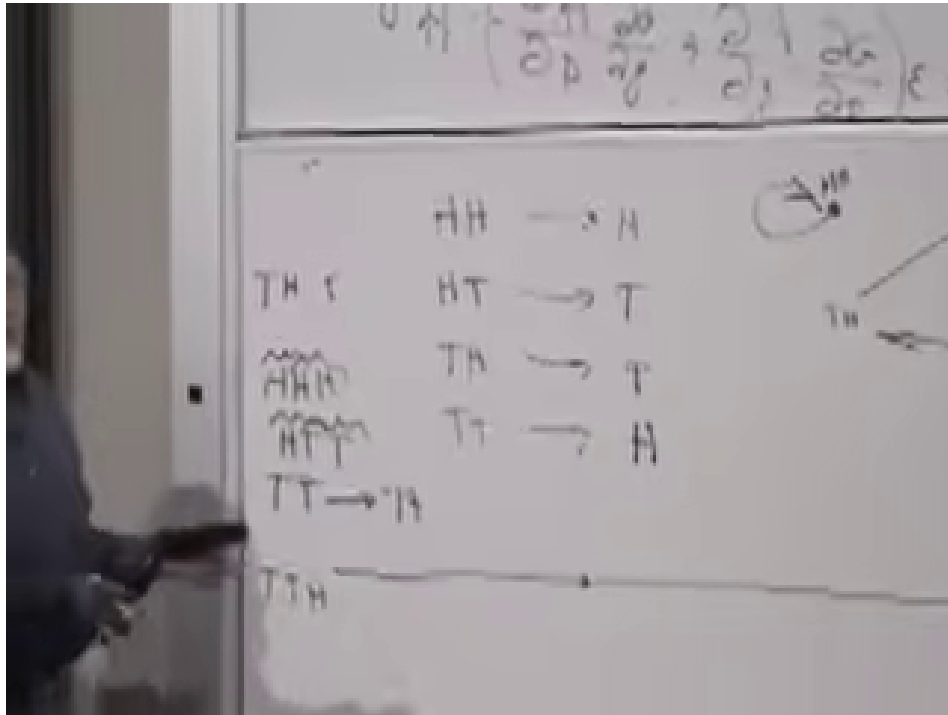


Figure 1.10: The board at 00:45:43 after the table has been corrected. The four pair states on the left generate a fixed point and a three-cycle on the right.

Lecture frame: 00:45:43.

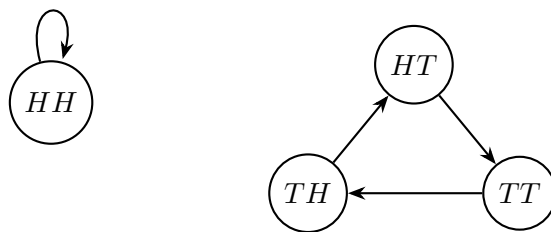


Figure 1.11: Clean reconstruction of the reversible four-state update in Eq. (1.28).

A student recognizes the output rule as exclusive OR if $H = 0$ and $T = 1$:

$$a_{n+2} = a_n + a_{n+1} \pmod{2}. \quad (1.29)$$

The pair update can then be written

$$\begin{pmatrix} a_{n+1} \\ a_{n+2} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} a_n \\ a_{n+1} \end{pmatrix} \pmod{2}. \quad (1.30)$$

Its matrix has nonzero determinant modulo two, so the pair update is invertible. The new symbol alone does not preserve the past, but the shifted pair does. Once again, the cure is not to abandon determinism; it is to identify the complete state.

Question from the audience. What if still more history is needed—for example, three symbols rather than two?^a

Response. Then the state must contain three symbols and the phase space contains 2^3 possible histories. The same principle applies in continuous mechanics. If position, velocity, and acceleration had to be specified independently, the fundamental equation would be third order and the state would be $(x, \dot{x}, \ddot{x})$. Experiment tells us that ordinary Newtonian mechanics is second order: position and velocity are enough.

^aLecture timestamp: 00:46:32–00:47:30.

1.11 What the Toy World Has Taught Us

The stroboscopic coin was not a diversion from mechanics. It isolated the structure that Newton's equations later realize. A state is all and only the information required to continue the motion. A classical closed-system law carries that information uniquely toward both the future and the past. On a finite phase space the law is a permutation; on a continuous phase space it is a flow.

For a particle, the state cannot be position alone. It must include velocity, and Newton's second-order equation is precisely what makes those two pieces of initial data necessary. Conservation laws, reversibility, the order of differential equations, and the geometry of phase space are therefore not separate topics. They are different expressions of one organizing principle: physical evolution must act on complete information.